

Project Plan — P02: Plant Disease Recognition from Leaf Images

Egor Savchenko Zamir Safin Ilia Ponomarev Vadim Poponnikov

Innopolis University — Computer Vision 2026

6 October 2026

1. Project question

Main question. Does full fine-tuning of an ImageNet-pretrained ResNet-50 retain its advantage over frozen-backbone linear probing when leaf images are degraded by realistic robotic field-capture artifacts rather than captured under laboratory conditions?

The controlled variable is which parameters receive gradients. Architecture, data, optimiser, schedule and the seed budget are shared; the learning rate is tuned per condition and every condition runs with the same three seeds, so a difference between them is attributable to the transfer strategy rather than to an untuned baseline or to initialisation luck.

Sub-questions.

- What is the clean-data gap between the two conditions, and at what training cost?
- How sensitive is the fine-tuned condition to the learning rate?
- How does each condition degrade under three independent capture artifacts at three severities?
- Which visually similar disease pairs are confused, and which classes fail first under degradation? (Required explicitly by the P02 brief.)

2. Dataset

PlantVillage, Kaggle mirror `mohitsingh1804/plantvillage`: 38 classes across 14 crops, laboratory-captured images of single detached leaves on a uniform background, 256×256 px. Chosen because it is public, class-balanced enough for macro-averaged metrics, and small enough that the whole study fits in the course scope without large-scale training.

Preprocessing: directory names are sanitised, images are resized to 224×224 and normalised with ImageNet statistics. Training augmentation is horizontal and vertical flips plus rotation up to 15° ; validation and test use no augmentation.

3. Train / validation / test split

Split	Images	Origin
Train	41 275	Kaggle <code>train</code> , minus the test sample
Validation	10 861	Kaggle <code>val</code> , used as shipped
Test	2 169	5 % of each training class, seed 42

Leakage control, and its limit. The test split is built by *moving* files out of `train`, never by copying, so no file appears in two splits. The split is fixed by a committed manifest of the 2 169 test files, and the script reproduces it by moving exactly those files, run once on a fresh download. Class lists are asserted identical across the splits, and model selection uses validation macro F1 only. The split is *not* grouped by leaf, however: PlantVillage holds several photographs of one physical leaf, so near-duplicates can fall on both sides. We measure that contamination with a perceptual hash and report its measured effect on accuracy rather than claiming there is none.

4. Evaluation measures

- **Accuracy** — headline measure, directly comparable with published PlantVillage results.
- **Macro F1** — primary measure for model selection and for the comparison, because it weights all 38 classes equally and is not dominated by the largest ones.
- **Per-class precision, recall and F1** — to locate which diseases fail.
- **Confusion pairs** — off-diagonal mass of the confusion matrix, to answer the visually-similar-disease question in the brief.
- **Expected calibration error** — calibration under distribution shift, binned over all predictions rather than over the handful that happen to be wrong.

- **Paired McNemar tests and Wilson intervals** — both conditions see byte-identical inputs, so the comparison is paired rather than a comparison of two proportions.
- **Cost** — trainable parameters, epochs to early stop, training time, inference latency at batch 1 and 64, and peak GPU memory.

5. Baseline and second condition

Baseline: ResNet-50 with ImageNet weights as a true linear probe — the backbone frozen *and* every BatchNorm module held in evaluation mode during training, so the features are genuinely fixed. Only the 38-way classifier is trained, 77 862 of 23 585 894 parameters.

Second condition: the same network with all 23 585 894 parameters trainable.

Learning rate: tuned separately for each condition on validation macro F1, since a linear probe on fixed features and full fine-tuning are different optimisation problems.

Seeds: three per condition, so that a difference can be separated from initialisation luck.

Degradation study: both trained models are evaluated, without retraining, on the clean test set and on degraded versions of it, modelling an autonomous field phenotyping robot — directional motion blur, low light with sensor noise and auto-gain, and JPEG compression, each at three severities. Because low light confounds darkening, noise and gain, a fourth condition is a control: additive Gaussian noise at unchanged brightness, matched to the noise level of the low-light condition.

6. Division of work

Member	Responsibility
Egor Savchenko	Overall architecture of the study and of the codebase. Formulation of the research question and of the experimental design. Implementation of the full pipeline: training in both conditions, the learning-rate sweeps, the seed replications, the capture-degradation model and its control, the inference benchmark, the failure and confusion analysis, and the results assembly. Testing and verification: recomputation of every reported metric from the raw per-image predictions, cross-checking of independent runs, and the reproducibility guarantees.
Zamir Safin	Dataset analysis and data handling: selection of the dataset and verification of its source and licence; inventory of the 38 classes across 14 crops and their per-class counts; construction of the held-out test split; perceptual-hash measurement of near-duplicate contamination and of its effect on the reported metrics; sanitisation of directory and file names; verification that the class lists of the three splits are identical; analysis of class balance and identification of under-represented classes; the preprocessing and augmentation policy; and the dataset provenance and download instructions in the README.
Iliia Ponomarev	Analysis of existing solutions and architectures: survey of transfer-learning approaches used for plant disease recognition; justification of ImageNet-pretrained ResNet-50 as the backbone; framing of frozen-backbone probing versus full fine-tuning as the controlled comparison; review of corruption and robustness benchmarking practice, which motivates the capture-degradation design; and positioning of our numbers against published PlantVillage baselines.
Vadim Poponnikov	Reporting, demo and presentation: the two-page technical summary, the presentation slides, and the live demo including its scenario, the failure case shown, and rehearsal against the time limit.

7. Schedule

Date	Milestone	Deliverable
6 Oct	Project plan	This document
13 Oct	First working model	Dataset pipeline and baseline results
20 Oct	Second approach	Fine-tuned condition and the comparison
27 Oct	Progress check	Preliminary comparison and remaining experiments
3 Nov	Results checkpoint	Final metrics, runtime, tables and figures, failure cases
10 Nov	Final submission	Two-page summary, repository, slides, demo, contributions